

Databricks APEX Deployment Runbook

Deploying to a new Databricks workspace (App + Bundle + Data Bootstrap)

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1) Scope

This runbook deploys the APEX app (`apex-etrm`) into another Databricks workspace, then bootstraps catalog/schemas and optional seed data.

2) Prerequisites

- Databricks CLI authenticated to target workspace profile
- Permissions to create apps, jobs, SQL objects, and secrets
- Warehouse ID available for bootstrap SQL
- External API keys (ENTSOE / ERCOT) if real ingestion is enabled

3) Authenticate and target workspace

Use a dedicated profile for the destination workspace.

```
databricks auth login https:// --profile
```

4) Create app (one-time)

If the app does not exist in the destination workspace:

```
databricks apps create apex-etrm -p
```

5) Upload source and deploy app

Project-mode app deploy can fail because of notebook path validation in `databricks.yml`. Use sync + source path deploy.

```
databricks sync . /Workspace/Users//apex-etrm -p  
databricks apps deploy apex-etrm --source-code-path /Workspace/Users//apex-etrm -p  
databricks apps get apex-etrm -p
```

6) Deploy bundle resources (jobs)

Deploy workflow jobs defined in `databricks.yml`.

```
databricks bundle deploy -t dev -p  
databricks jobs list -p
```

7) Bootstrap catalog and schemas

Primary catalog is `apex\_fresh`. Bootstrap options:

- Option A: Run SQL setup files from `sql/setup/` in order
- Option B: Run `scripts/setup\_database.py` (Databricks SDK executor)
- Option C: Use helper `scripts/run\_sql\_files.py` for statement API execution

Recommended bootstrap command:

```
DATABRICKS_SQL_WAREHOUSE_ID= python3 scripts/setup_database.py
```

SQL execution order:

- `sql/setup/00\_create\_catalog.sql`
- `sql/setup/01\_create\_schemas.sql`
- `sql/setup/02\_create\_market\_tables.sql`
- `sql/setup/03\_create\_trading\_tables.sql`
- `sql/setup/04\_create\_risk\_tables.sql`
- `sql/setup/05\_create\_portfolio\_tables.sql`
- `sql/setup/06\_create\_analytics\_tables.sql`

8) Configure app environment and secrets

Variable	Purpose
DATABRICKS_SQL_WAREHOUSE_ID	Warehouse used by runtime queries
APEX_CATALOG	Catalog name (default: apex_fresh)
DATABRICKS_WORKSPACE_URL	Workspace URL secret reference
LAKEBASE_HOST	Lakebase/Postgres host secret
LAKEBASE_DATABASE	Lakebase/Postgres database secret

9) Post-deploy validation

- App status is RUNNING and URL is reachable
- APEX API health endpoint returns healthy state
- Catalog exists and schemas/tables are queryable
- Ingestion jobs exist and can run-now successfully

10) Rollback and recovery

If deployment fails, redeploy previous workspace snapshot or stop app and redeploy known-good commit via sync + apps deploy.

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